Summary in Graph

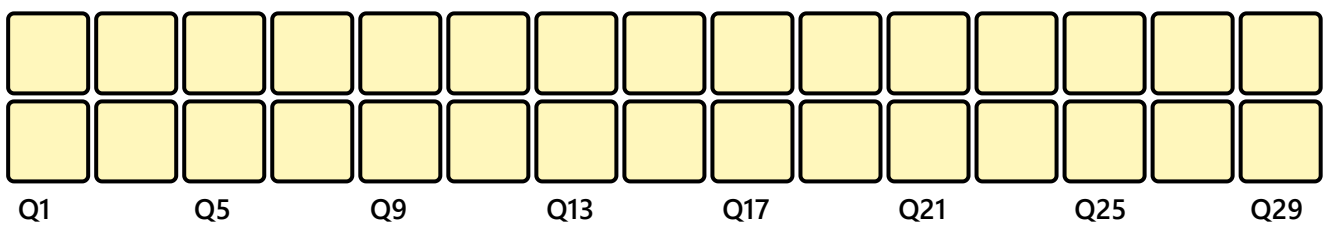
Exam Summary (GO Classes CS Test Series 2026 | Data Structures | Subject Wise Test 3)

Qs. Attempted:	0 0 + 0	Correct Marks:	0 0 + 0
Correct Attempts:	0 0 + 0	Penalty Marks:	0 0 + 0
Incorrect Attempts:	0 0 + 0	Resultant Marks:	0 0 + 0

Total Questions:	30 20 + 10
Total Marks:	40 20 + 20
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (36) out of 359 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

You are given a circular queue CQ with a capacity of 8 elements. Initially, the queue is empty. The following sequence of operations is performed on CQ in order:

$\text{enqueue}(6);\ \text{enqueue}(5);\ \text{enqueue}(3);\ \text{enqueue}(2);\ \text{dequeue}();\ \text{dequeue}();$

```
  0    1    2    3    4    5    6    7
-----------------------------------------
|    |    | 3  | 2  |    |    |    |    |
-----------------------------------------
             ^    ^
           front back
```

After these operations, the queue receives an additional sequence of operations, executed in order:

$\text{enqueue}(7); \text{enqueue}(8); \text{enqueue}(2); \text{enqueue}(11); \text{dequeue}(); \text{dequeue}(); \text{enqueue}(9);$

What will be the final positions (indices) of the front and rear pointers after the completion of all operations?

A. Front $= 0$ and Rear $= 7$

B. Front $= 2$ and Rear $= 9$

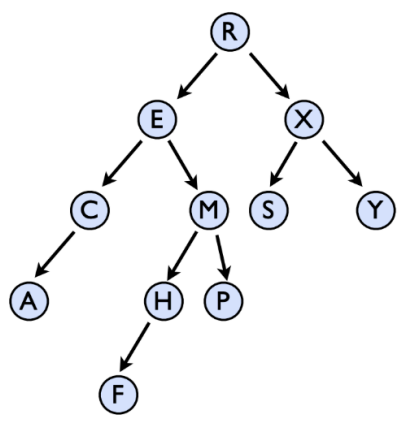
C. Front $= 4$ and Rear $= 0$

D. Front $= 3$ and Rear $= 7$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider the imbalanced Adelson-Velsky Landis(AVL) Tree in the figure below.



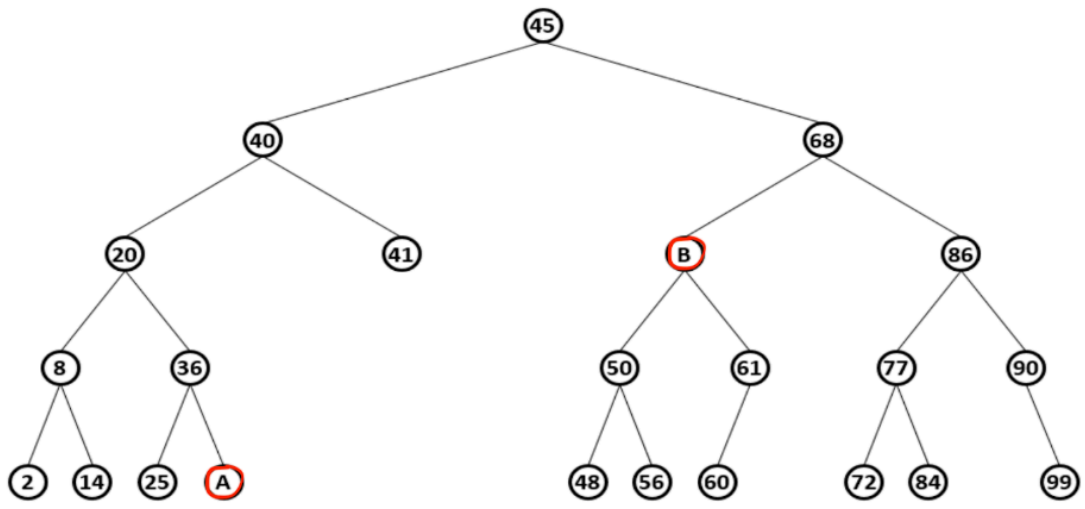
If the last inserted node is F in the AVL tree then which of the following is the correct imbalance and preorder traversal after rotation?

A. Left-Right imbalance
B. Left-Left imbalance
C. $\text{M E C A H F R P X S Y}$
D. $\text{R E C A H G P M X S Y}$

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

The graph shown below is a Binary Search Tree(BST):

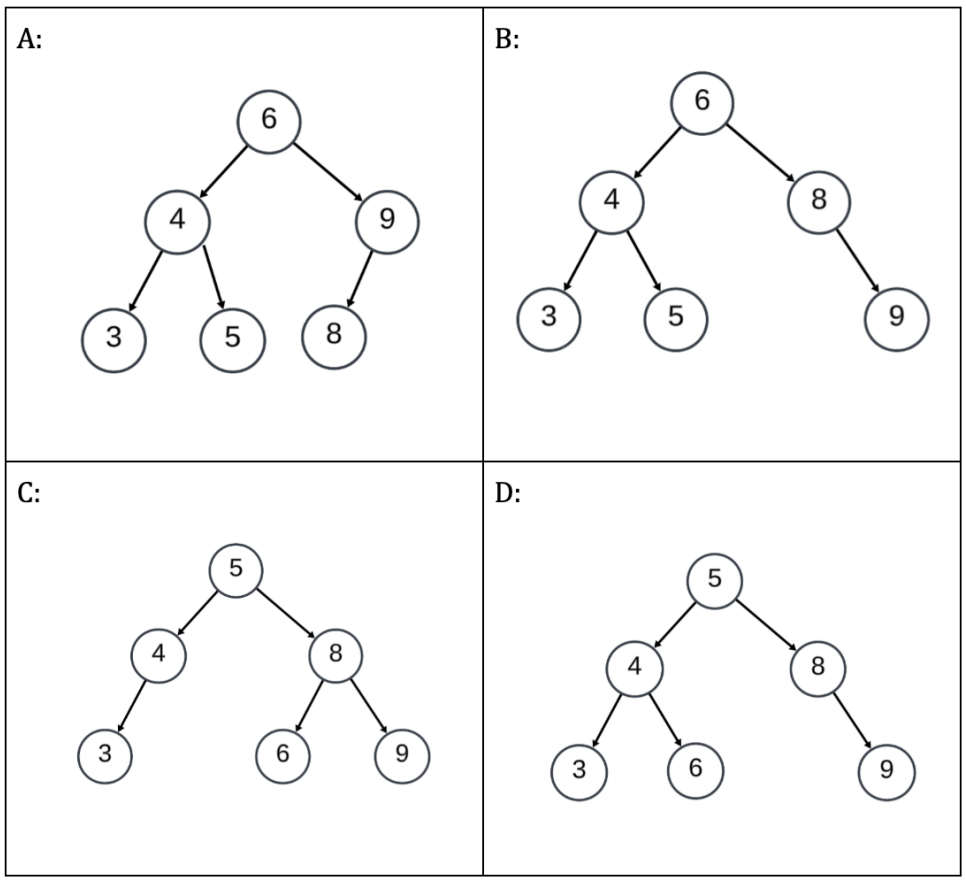


For the missing values of nodes A and B what will be the maximum possible difference?

Your Answer: | Correct Answer: 24 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Which of the following is correct AVL tree, constructed by inserting the keys $4, 8, 9, 3, 5, 6$ in that order:



Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Consider the following expression tree,

```
          *
        /   \
       +     -
      / \   / \
     +   7 8   *
    / \       / \
   1   2     4   +
                / \
               1   5
```

Which of the following is the correct postfix and prefix translation of the expression represented by the above

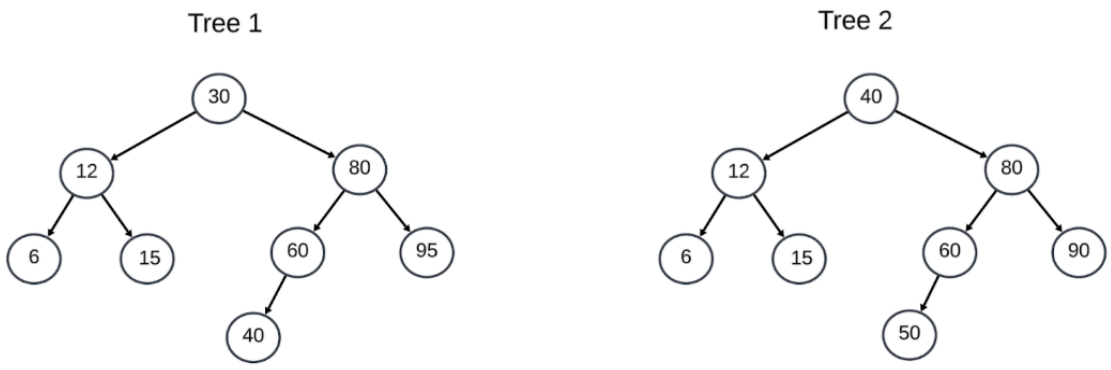
tree?

A. Prefix: $* + +127 - 8 * 4 + 15$
B. Postfix: $12 + 7 + 8 * -4 + *15+$
C. Prefix: $* + +12 - * + 1548$
D. Postfix: $12 + 7 + 8415 + * - *$

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following height balanced Binary Search Trees Tree 1 and Tree 2.



Which of the following sequence of operations can convert Tree1 to Tree2?

A. delete(30); delete(95); insert(90); insert(50);
B. delete(95); insert(95); insert(50); delete(30);
C. delete(30); insert(50); delete(95); insert(90);
D. delete(95); insert(50); delete(30); insert(90);

Your Answer: | Correct Answer: A;C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

Which of the following statement(s) is TRUE about a binary heap of n nodes?

A. Two binary heaps can be merged into one in $\mathrm{O}(1)$ time.
B. A binary heap can be converted into a Binary Search Tree (not necessarily balanced) in $\mathrm{O}(\mathrm{n})$ time.
C. Insertion in a binary heap requires $\mathrm{O}(\mathrm{logn})$ time.
D. Deletion of the i-th node in a binary heap requires $\mathrm{O}(\mathrm{n})$ time.

Your Answer: | Correct Answer: C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider that a heap of height h has k entries in the last level. How many nodes does the heap have in total?

A. $2^h + k$
B. $2^h - 1 + k$
C. $2^{h-1} + k$
D. $2^{h-1} - 1 + k$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

Suppose that you implement a queue using a null-terminated singly-linked list, maintaining a reference to the item least recently added but not maintaining a reference to the item most recently added. What are the worst-case running times for enqueue and dequeue?

A. Constant time for both enqueue and dequeue.
B. Constant time for enqueue and linear time for dequeue.
C. Linear time for enqueue and constant time for dequeue.
D. Linear time for both enqueue and dequeue.

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #10 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

Consider the bottom-up construction of an array-based heap. What happens after the first phase in which all entries are added to the array?

A. Down-heap bubbling from the root node.
B. Down-heap bubbling from all nodes except the ones in the last level, starting from the root node and ending at the last node in the last but one level.
C. Down-heap bubbling from all nodes except the ones in the last level, starting from the last node of the level before the last level and ending up at the root node.
D. Up-heap bubbling from the root node.

Your Answer: Correct Answer: C Not Attempted Time taken: 00min 00sec Discuss

Q #11 Multiple Select Type Award: 1 Penalty: 0 Data Structures

Which of the following statement(s) is FALSE about a binary heap of distinct keys?

A. With the build heap method, an array with distinct integers can be sorted in $\mathrm{O}(\mathrm{n})$ time.
B. To delete the i-th node in a min-heap, exchange the last node with the i-th node, then do the min-heapify on the i-th node.
C. In a max-heap, the next largest element of a key can be found either on the same level or on at most one level above(or below) to the current level.
D. In a binary min-heap, the maximum element is always found on the last level.

Your Answer: Correct Answer: A;B;C;D Not Attempted Time taken: 00min 00sec Discuss

Q #12 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

Suppose a hash table has size 10, and that the search keys are strings consisting of 3 lower case letters. We want to hash 7 unknown values from this keyspace. In the hash function, when we refer to the alphabet positions of the letters, we mean: " a " $= 1$, " b " $= 2, \ldots,$ " z " $= 26$.
$h(k) =$(product of the alphabet positions of letters) $\%10$
Which of these hash function characteristics are violated by this hash function?

I. A hash function distributes the keys uniformly over the array.
II. A hash function is deterministic(produces exact results every time computed for any key).
III. A hash function is computed in constant time.

A. Only (I) is violated.

B. Only (III) is violated.
C. At least two of (II), (II) and (III) are violated.
D. None of these characteristics are violated-our hash function is a good one!

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Multiple Select Type | Award: 1 | Penalty: 0 | Data Structures

A linear-probing hash table of length 10 uses the hash function $\mathrm{h(k)} = \mathrm{k} \bmod 10$ to hash integer keys between 0 and 100 . After inserting 8 keys into an empty hash table, the table is as shown below.

0	1	2	3	4	5	6	7	8	9
19	91	21	93	74	2		77		89

Which of the following choices gives a possible order in which the keys could have been inserted into an initially empty hash table? Assume that the initial length of the array is 10 and that it neither grows nor shrinks.

A. $19, 91, 21, 93, 77, 89, 74, 2$
B. $77, 89, 19, 93, 91, 74, 21, 2$
C. $89, 19, 93, 91, 74, 21, 2, 77$
D. $91, 77, 93, 21, 89, 74, 2, 19$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Assume that `LL` is a DOUBLY linked list with the head node and at least one other internal node which is not the last node.

`head.value`: dereference the value at node pointed by head.
`head.next:` reference to the next node pointed by head.
`head.prev`: reference to the previous node pointed by head.

```
Procedure(head, X):
    While ( head.value  !=  X ):
        head = head.next

    temp = head
    head = head.next
    head.prev = temp.prev;
    temp.prev.next = head;
    delete temp
```

Analyze the above pseudocode procedure called with parameters head of `LL` and a value X. Which of the following is TRUE about the procedure?

A. Deletes the immediate first node after the first occurrence of node with value X.
B. Swaps the node X value with its immediate predecessor node.
C. Deletes the first occurrence of node with value X.
D. Deletes every occurrence of node with value X.

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

Consider the following sequence of operations, on an initially empty stack S:
{push(5), push(3), pop(), push(2), push(8), pop(), S.size(), push(1), pop(), pop(), pop()}.
Which sequence of values is returned by this sequence of operations?

A. $\{-, -, 3, -, -, 8, 3, -, 1, 2, 5\}$
B. $\{-, -, 3, -, -, 8, 2, -, 1, 2, 5\}$
C. $\{-, -, 3, -, -, 8, 2, -, 1, 3, 5\}$
D. $\{-, -, 5, -, -, 3, 2, -, 2, 8, 1\}$

Your Answer: Correct Answer: B Not Attempted Time taken: 00min 00sec Discuss

Q #16 Multiple Select Type Award: 1 Penalty: 0 Data Structures

In a queue implemented with a circular array of size N, where head and tail are associated with the indices of the next element to dequeue and the most recently enqueued element, respectively, which of the following expressions evaluates to True if and only if the queue is full?

A. $(\text{tail} + 1)\ \%\ \text{N} == \text{head}$
B. $(\text{head} + 1)\ \%\ \text{N} == \text{tail}$
C. $\text{tail}\ \%\ \text{N} == \text{head - 1}$
D. $(\text{head} + \text{N})\ \%\ \text{N} == \text{head}$

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 02sec Discuss

Q #17 Multiple Choice Type Award: 1 Penalty: 0.33 Data Structures

What are the contents of the list L after executing the following code?

```
Initialize an empty stack S
Initialize an empty queue Q
Initialize an empty list L

for x from 0 to 9:
    S.push(x)

while S is not empty:
    element = S.pop()
    Q.enqueue(element)

while Q is not empty:
    element = Q.dequeue()
    L.append(element)
```

A. $[9, 8, 7, 6, 5, 4, 3, 2, 1, 0]$
B. $[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]$
C. $[1, 0, 3, 2, 5, 4, 7, 6, 9, 8]$
D. $[]$

Your Answer: Correct Answer: A Not Attempted Time taken: 00min 00sec Discuss

Q #18 Multiple Select Type Award: 1 Penalty: 0 Data Structures

Suppose that a person performs an intermixed sequence of enqueue and dequeue operations such that enqueues and dequeues could occur in any order. For example, one enqueue operation and the rest all dequeue operations, or all operations of one type, or a blend of enqueue and dequeue in random order.

- The enqueue operations add the integers 0 through 9 in order onto the queue.
- The dequeue operations remove the integers from the queue and print out the return Value.

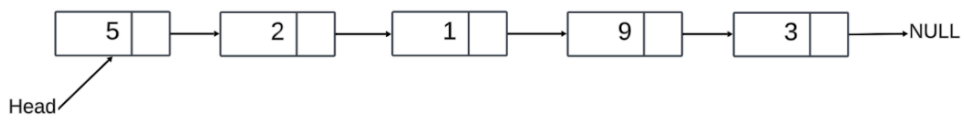
Which of the following output sequence(s) could not occur?

A. 6541032789
B. 1234569870
C. 1345678902
D. 1236587904

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Data Structures

Consider the following singly linked list $\mathtt{LL}$:




```
Node *p, *addNode;
p = head → next;
addNode = createNode(7);
addNode → next = p → next → next;
P → next = addNode;
```

Where:
$\mathtt{Head}$: points to the first node of the linked list.
$\mathtt{createNode(x)}$: dynamically creates a node, sets the next field NULL and returns the address.
$\mathtt{Node} \rightarrow \mathtt{next}$ references to the next node of the linked list.
If we execute the above code on the singly linked list $\mathtt{LL}$ then then what will be the final linked list?

A. $5 \rightarrow 2 \rightarrow 7 \rightarrow 1 \rightarrow 9 \rightarrow 3$
B. $5 \rightarrow 2 \rightarrow 7 \rightarrow 9 \rightarrow 3$
C. $5 \rightarrow 2 \rightarrow 1 \rightarrow 9 \rightarrow 3$
D. $5 \rightarrow 7 \rightarrow 2 \rightarrow 1 \rightarrow 9 \rightarrow 3$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Numerical Type | Award: 1 | Penalty: 0 | Data Structures

Suppose that we have a hash table with $2n$ slots, with collisions resolved by chaining, and suppose that n keys are inserted into the table. Assume simple uniform hashing, i.e., each key is equally likely to be hashed into each slot. What is the expected number of elements that hash into slot i?

Your Answer: | Correct Answer: 0.5 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Suppose that we have a hash table with $2n$ slots, with collisions resolved by chaining, and suppose that n keys are inserted into the table. Assume simple uniform hashing, i.e., each key is equally likely to be hashed into each slot.
What is the probability that exactly k keys hash into slot i?

A. $\binom{n}{k}\left(\frac{1}{2n}\right)^k\left(\frac{2n-1}{2n}\right)^{n-k}$
B. $\binom{n}{k}\left(\frac{1}{2n}\right)^k\left(\frac{k}{2n}\right)^{n-k}$
C. $\binom{n}{k}\left(\frac{1}{2n}\right)^k\left(\frac{1}{2n}\right)^{n-k}$
D. $\binom{n}{k}\left(\frac{1}{2n}\right)^k\left(\frac{2n-k}{2n}\right)^{n-k}$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Chaining and probing are two methods used to resolve collisions in hash tables so that the amortized access time is $\mathrm{O}(1)$. For each of the following claims, indicate whether it is true of chaining, probing, both, or neither.

I. Needs additional memory beyond the primary array for the hash table.
II. May have either "linear" or "quadratic " hash functions.
III. Crashes if the load factor becomes greater than 1.
IV. Requires computing the hash function multiple times when doing an insertion.

A. Chaining, probing, probing, neither
B. Chaining, both, probing, probing
C. Chaining, probing, both, probing
D. Chaining, probing, both, neither

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Suppose that we have numbers between 1 and 1000 in a binary search tree and we want to search for the number 363. Which of the following sequences cannot be the sequence of nodes visited in the search?

A. $2, 252, 401, 398, 330, 344, 397, 363$
B. $924, 220, 911, 244, 898, 258, 362, 363$
C. $2, 399, 387, 219, 266, 382, 381, 278, 363$
D. $925, 202, 911, 240, 912, 245, 363$

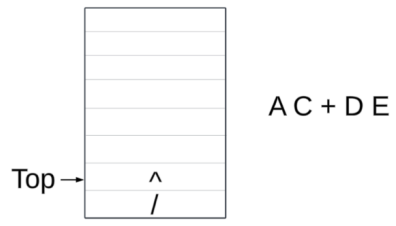
Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

A Stack is used to convert infix expression $(\mathrm{A}+\mathrm{C})/\mathrm{D}^{\wedge}\mathrm{E}^{*}\ \mathrm{B}$ to postfix expression according to following table,

Operator	Precedence	Associativity
$\wedge$	Highest	Right
$/$ and $*$	Middle	Right
$+$ and $-$	Lowest	Left

An intermediate state of stack and postfix expression is given below



What will be the content (starting from bottom to top) of the stack immediately after the above state while conversion?

A. $*$
B. $/*$
C. $/^\wedge$
D. $/^\wedge *$

Your Answer: | Correct Answer: B | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

A binary tree is a rooted tree in which each node has at most two children. If such a binary tree has height h (defined as the number of edges on the longest root-to-leaf path) and contains n nodes, then which of the following inequalities always holds?

A. $\log\left(\frac{n+1}{2}\right) <= h <= n-1$
B. $\log\left(\frac{n}{2}\right) <= h <= n-1$
C. $\log\left(\frac{n}{2}\right) <= h <= 1$
D. $\log\left(\frac{n+1}{2}\right) <= h <= 1$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 | Numerical Type | Award: 2 | Penalty: 0 | Data Structures

Consider a Binary Search Tree (BST) constructed with 1000 distinct keys, where the keys are inserted one by one into an initially empty tree. What will be the minimum number of comparisons made during the construction of the tree?

Your Answer: | Correct Answer: 7987 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Consider a Binary Search Tree T with the `root` node.
The following recursive pseudocode is defined:

```
Void mystery(root){
    if( root == NULL ) return;

    if( root.right != NULL){
        return mystery(root.right);
    }

    print(root.val);
}
```

where:
`root`: reference to the root node of the tree.
`Node.right`: reference to the right child of the node.

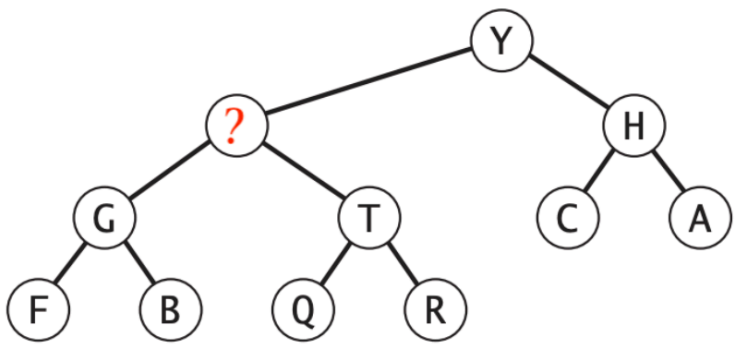
Which of the following is TRUE about the above pseudocode?

A. The call `mystery(root.right)` prints the maximum value node in the tree T.
B. The call `mystery(root)` prints the maximum value node in the tree T.
C. The call `mystery(root.left)` prints the maximum value node in the left subtree of the tree T.
D. The call `mystery(root.left)` prints the inorder predecessor of the `root` node.

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 | Multiple Select Type | Award: 2 | Penalty: 0 | Data Structures

Consider the following max-heap tree of distinct keys:



If the last inserted key is 'Y', then which of the following keys can be the node with the question mark?

A. U
B. V
C. W
D. X

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Suppose we use a hash function to hash n distinct keys into an array of length m. Assuming simple uniform hashing, the expected number of colliding pairs of elements is _____ ?

A. $\theta\left(\frac{n}{m}\right)$
B. $\theta\left(\frac{n \cdot \log n}{m}\right)$
C. $\theta\left(\frac{n^2}{m}\right)$
D. $\theta\left(m^n\right)$

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Data Structures

Assuming uniform hashing, What is the probability of there being no collisions in a hashtable with 100 buckets and 3 total entries?

A. $99/100 * 98/100$

B. $1 - 99/100 * 98/100 * 97/100$

C. $1/100 * 1/100 * 1/100$

D. $1 - 1/100 * 1/100 * 1/100$

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

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